

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [4]: #loading the datasets

orders = pd.read_csv("List of Orders.csv")
details = pd.read_csv("Order Details.csv")
targets=pd.read_csv("Sales target.csv")

print(orders.head())
print(details.head())
print(targets.head())
```

	Order ID	Order Date	CustomerName	State	City
0	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
1	B-25602	01-04-2018	Pearl	Maharashtra	Pune
2	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
3	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
4	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata

	Order ID	Amount	Profit	Quantity	Category	Sub-Category
0	B-25601	1275.0	-1148.0	7	Furniture	Bookcases
1	B-25601	66.0	-12.0	5	Clothing	Stole
2	B-25601	8.0	-2.0	3	Clothing	Hankerchief
3	B-25601	80.0	-56.0	4	Electronics	Electronic Games
4	B-25602	168.0	-111.0	2	Electronics	Phones

	Month of Order Date	Category	Target
0	Apr-18	Furniture	10400.0
1	May-18	Furniture	10500.0
2	Jun-18	Furniture	10600.0
3	Jul-18	Furniture	10800.0
4	Aug-18	Furniture	10900.0

```
In [5]: print(orders.info())
print(details.info())
print(targets.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 560 entries, 0 to 559
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Order ID        500 non-null    object
1   Order Date      500 non-null    object
2   CustomerName    500 non-null    object
3   State           500 non-null    object
4   City            500 non-null    object
dtypes: object(5)
memory usage: 22.0+ KB
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Order ID        1500 non-null    object
1   Amount          1500 non-null    float64
2   Profit          1500 non-null    float64
3   Quantity        1500 non-null    int64
4   Category        1500 non-null    object
5   Sub-Category    1500 non-null    object
dtypes: float64(2), int64(1), object(3)
memory usage: 70.4+ KB
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36 entries, 0 to 35
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Month of Order Date  36 non-null    object
1   Category         36 non-null    object
2   Target           36 non-null    float64
dtypes: float64(1), object(2)
memory usage: 996.0+ bytes
None

```

```

In [6]: print(orders.isnull().sum())
        print(details.isnull().sum())

```

```

Order ID        60
Order Date      60
CustomerName    60
State           60
City            60
dtype: int64
Order ID        0
Amount          0
Profit          0
Quantity        0
Category        0
Sub-Category    0
dtype: int64

```

```

In [7]: print(orders.duplicated().sum())
        print(details.duplicated().sum())

```

59
0

```
In [13]: print(orders['Order Date'].head())
```

```
0    01-04-2018
1    01-04-2018
2    03-04-2018
3    03-04-2018
4    05-04-2018
Name: Order Date, dtype: object
```

```
In [15]: # date conversion
```

```
orders['Order Date'] = pd.to_datetime(orders['Order Date'], dayfirst=True)
```

```
In [17]: print(orders['Order ID'].nunique())
print(details['Order ID'].nunique())
```

```
500
500
```

```
In [19]: orders = orders.dropna()
details = details.dropna()

orders = orders.drop_duplicates()
details = details.drop_duplicates()
```

```
In [21]: df = pd.merge(orders, details, on='Order ID', how='inner')
```

```
In [23]: print(df.shape)
print(df.head())
```

```
(1500, 10)
   Order ID Order Date CustomerName      State      City  Amount  Profit \
0  B-25601  2018-04-01        Bharat    Gujarat  Ahmedabad   1275.0  -1148.0
1  B-25601  2018-04-01        Bharat    Gujarat  Ahmedabad    66.0   -12.0
2  B-25601  2018-04-01        Bharat    Gujarat  Ahmedabad     8.0    -2.0
3  B-25601  2018-04-01        Bharat    Gujarat  Ahmedabad    80.0   -56.0
4  B-25602  2018-04-01         Pearl  Maharashtra      Pune   168.0  -111.0

   Quantity  Category      Sub-Category
0          7  Furniture        Bookcases
1          5  Clothing                Stole
2          3  Clothing        Hankerchief
3          4  Electronics  Electronic Games
4          2  Electronics                Phones
```

```
In [25]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              1500 non-null   object
1   Order Date            1500 non-null   datetime64[ns]
2   CustomerName          1500 non-null   object
3   State                 1500 non-null   object
4   City                  1500 non-null   object
5   Amount                1500 non-null   float64
6   Profit                1500 non-null   float64
7   Quantity              1500 non-null   int64
8   Category              1500 non-null   object
9   Sub-Category          1500 non-null   object
dtypes: datetime64[ns](1), float64(2), int64(1), object(6)
memory usage: 117.3+ KB

```

```
In [27]: df.describe()
```

```
Out[27]:
```

	Order Date	Amount	Profit	Quantity
count	1500	1500.000000	1500.000000	1500.000000
mean	2018-10-23 14:00:57.600000	287.668000	15.970000	3.743333
min	2018-04-01 00:00:00	4.000000	-1981.000000	1.000000
25%	2018-07-26 18:00:00	45.000000	-9.250000	2.000000
50%	2018-11-08 00:00:00	118.000000	9.000000	3.000000
75%	2019-01-27 00:00:00	322.000000	38.000000	5.000000
max	2019-03-31 00:00:00	5729.000000	1698.000000	14.000000
std	NaN	461.050488	169.140565	2.184942

```
In [29]: print(df['Order ID'].nunique())
print(df['CustomerName'].nunique())
```

```
500
332
```

```
In [31]: df['customer_name_clean'] = (
    df['CustomerName']
    .str.strip()           # remove spaces at start/end
    .str.lower()           # make lowercase
    .str.replace(r'\s+', ' ', regex=True) # remove extra spaces
)
```

```
In [33]: print("Original unique names:", df['CustomerName'].nunique())
print("Cleaned unique names:", df['customer_name_clean'].nunique())
```

```
Original unique names: 332
Cleaned unique names: 332
```

```
In [35]: df['Customer_id'] = df['CustomerName'].astype('category').cat.codes
```

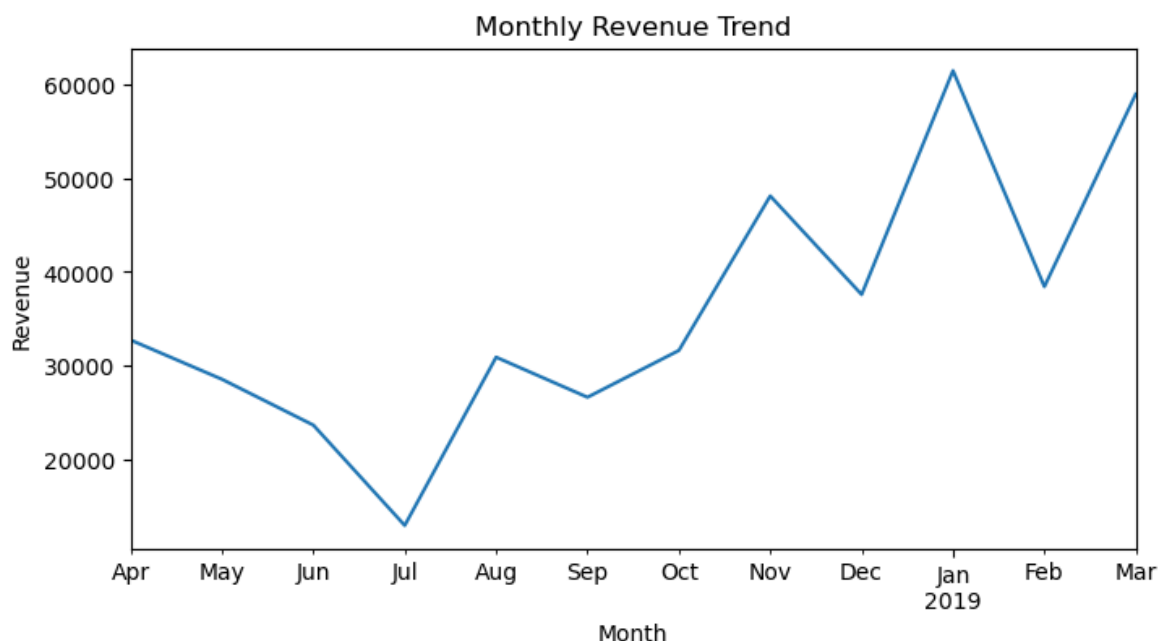
```
In [37]: df['year'] = df['Order Date'].dt.year
df['month'] = df['Order Date'].dt.month
```

```
df['year_month'] = df['Order Date'].dt.to_period('M')
```

```
In [39]: df.info()
```

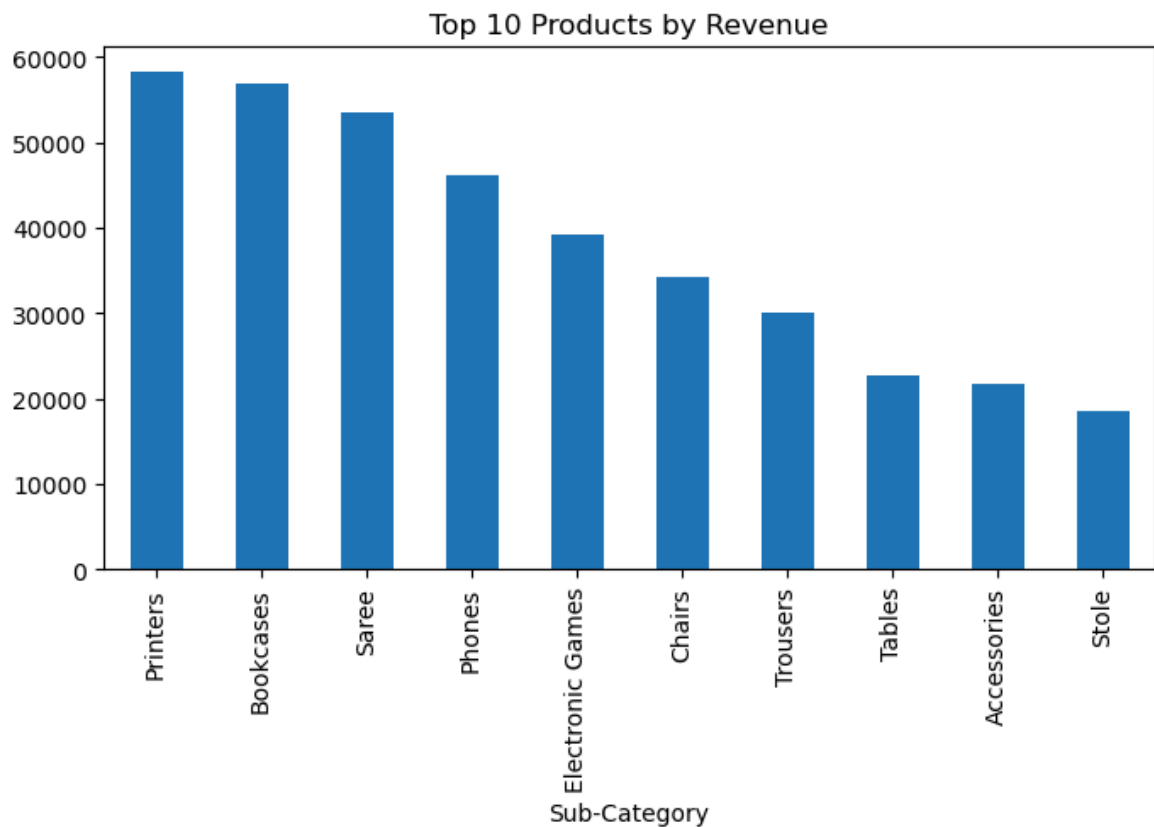
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Order ID              1500 non-null   object
1   Order Date            1500 non-null   datetime64[ns]
2   CustomerName          1500 non-null   object
3   State                 1500 non-null   object
4   City                  1500 non-null   object
5   Amount                1500 non-null   float64
6   Profit                1500 non-null   float64
7   Quantity              1500 non-null   int64
8   Category              1500 non-null   object
9   Sub-Category          1500 non-null   object
10  customer_name_clean    1500 non-null   object
11  Customer_id            1500 non-null   int16
12  year                   1500 non-null   int32
13  month                  1500 non-null   int32
14  year_month             1500 non-null   period[M]
dtypes: datetime64[ns](1), float64(2), int16(1), int32(2), int64(1), object(7), p
eriod[M](1)
memory usage: 155.4+ KB
```

```
In [41]: monthly_revenue = df.groupby('year_month')['Amount'].sum()
plt.figure(figsize=(8,4))
monthly_revenue.plot()
plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```

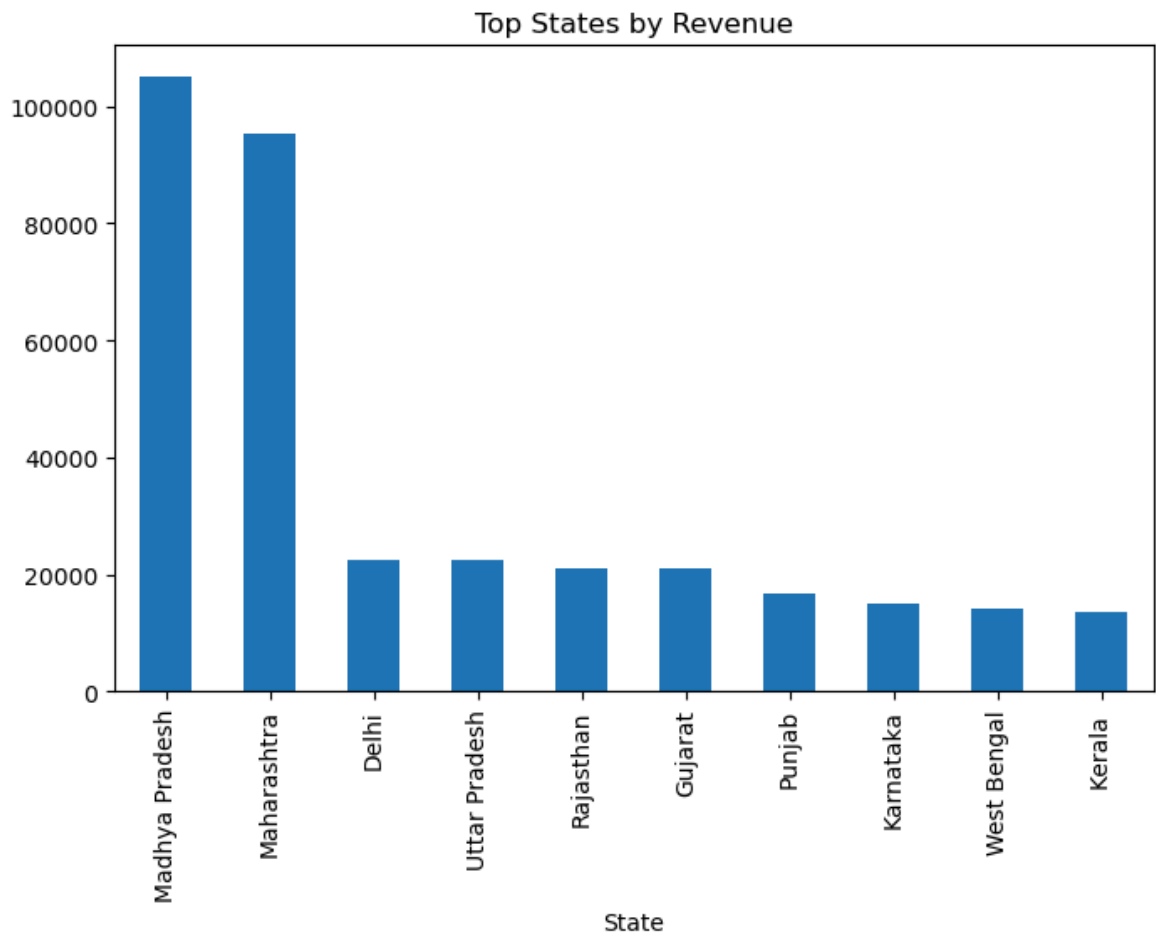


```
In [43]: top_products = df.groupby('Sub-Category')['Amount'] \
        .sum() \
        .sort_values(ascending=False) \
        .head(10)
```

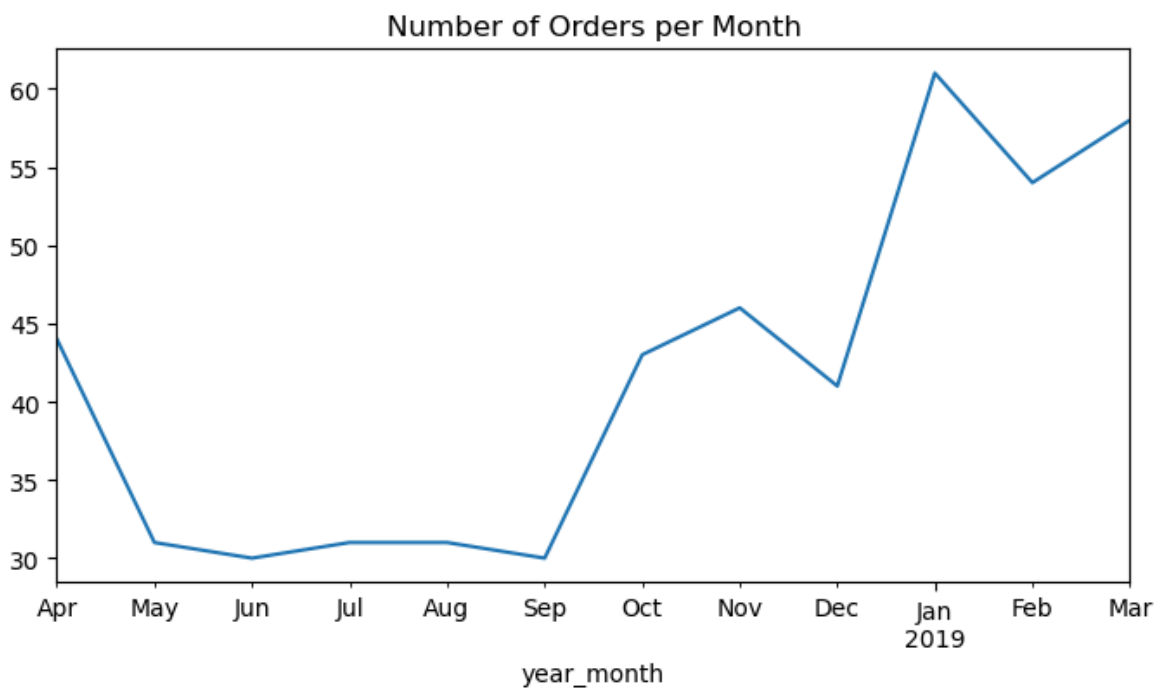
```
plt.figure(figsize=(8,4))
top_products.plot(kind='bar')
plt.title("Top 10 Products by Revenue")
plt.show()
```



```
In [45]: country_sales = df.groupby('State')['Amount'] \
        .sum() \
        .sort_values(ascending=False) \
        .head(10)
plt.figure(figsize=(8,5))
country_sales.plot(kind='bar')
plt.title("Top States by Revenue")
plt.show()
```

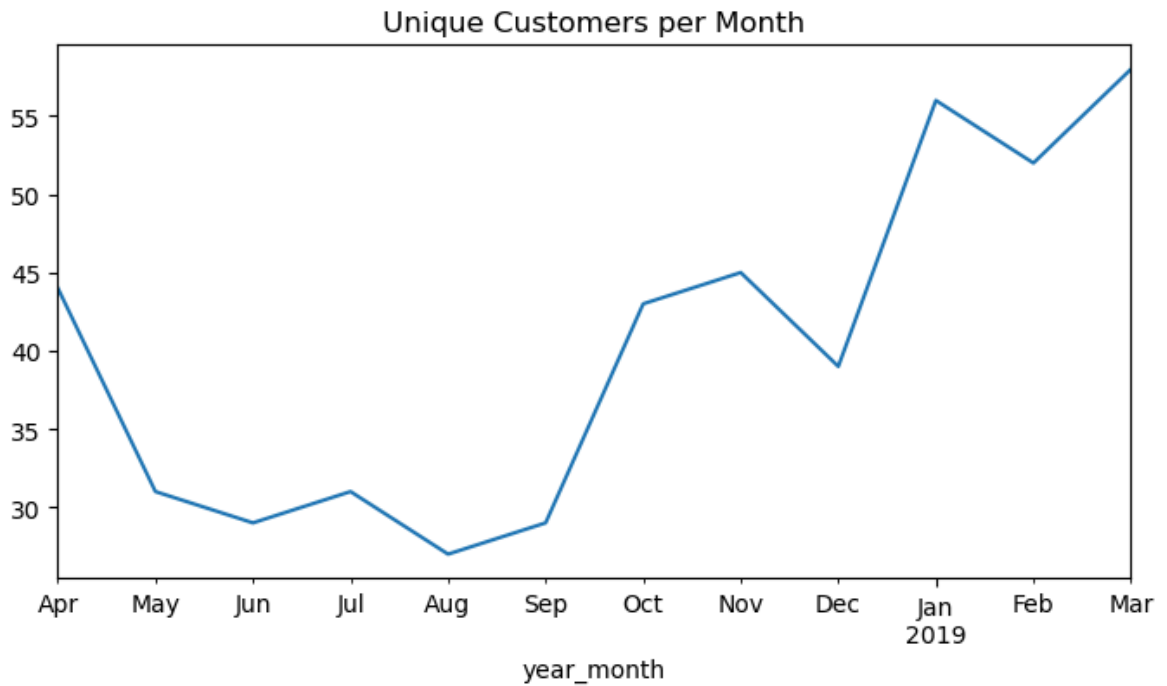


```
In [47]: orders = df.groupby('year_month')['Order ID'].nunique()  
plt.figure(figsize=(8,4))  
orders.plot()  
plt.title("Number of Orders per Month")  
plt.show()
```

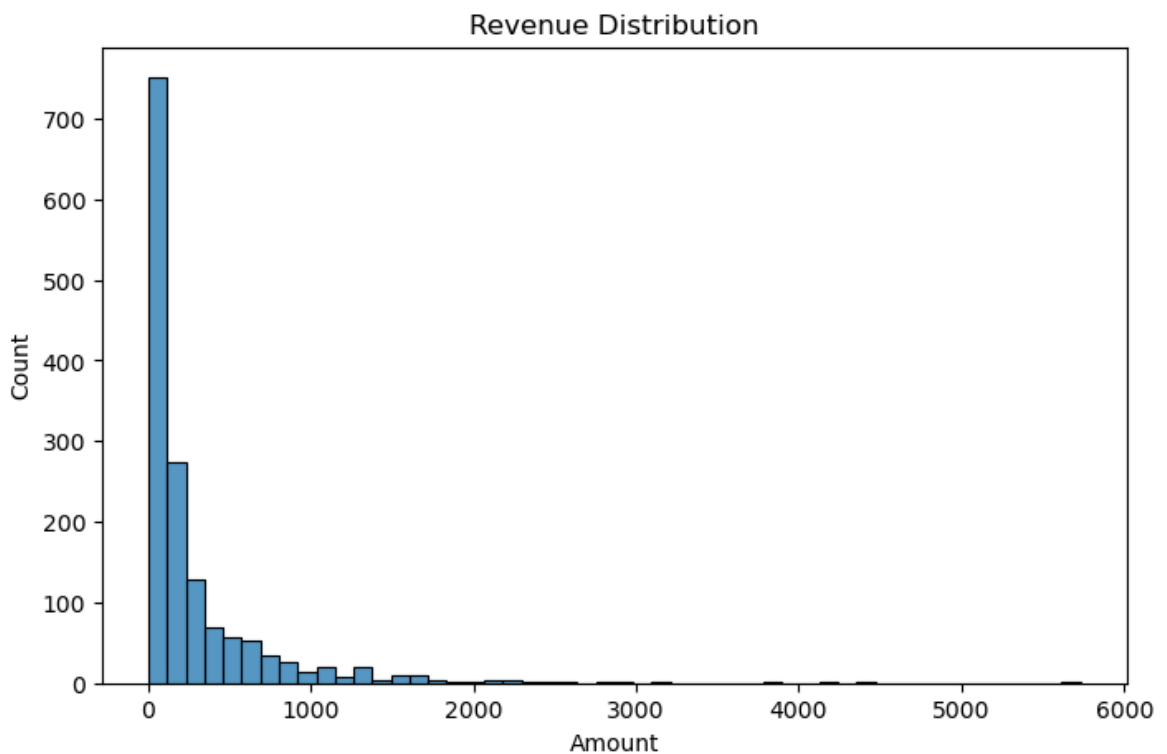


```
In [49]: customers = df.groupby('year_month')['Customer_id'].nunique()  
plt.figure(figsize=(8,4))  
customers.plot()
```

```
plt.title("Unique Customers per Month")
plt.show()
```



```
In [51]: plt.figure(figsize=(8,5))
sns.histplot(df['Amount'], bins=50)
plt.title("Revenue Distribution")
plt.show()
```



```
In [53]: df['cohort_month'] = df.groupby('CustomerName')['Order Date'] \
        .transform('min') \
        .dt.to_period('M')
```

```
In [55]: df['OrderMonth'] = df['Order Date'].dt.to_period('M')
```

```
df['CohortIndex'] = (df['OrderMonth'] - df['cohort_month']).apply(lambda x: x.n)
```

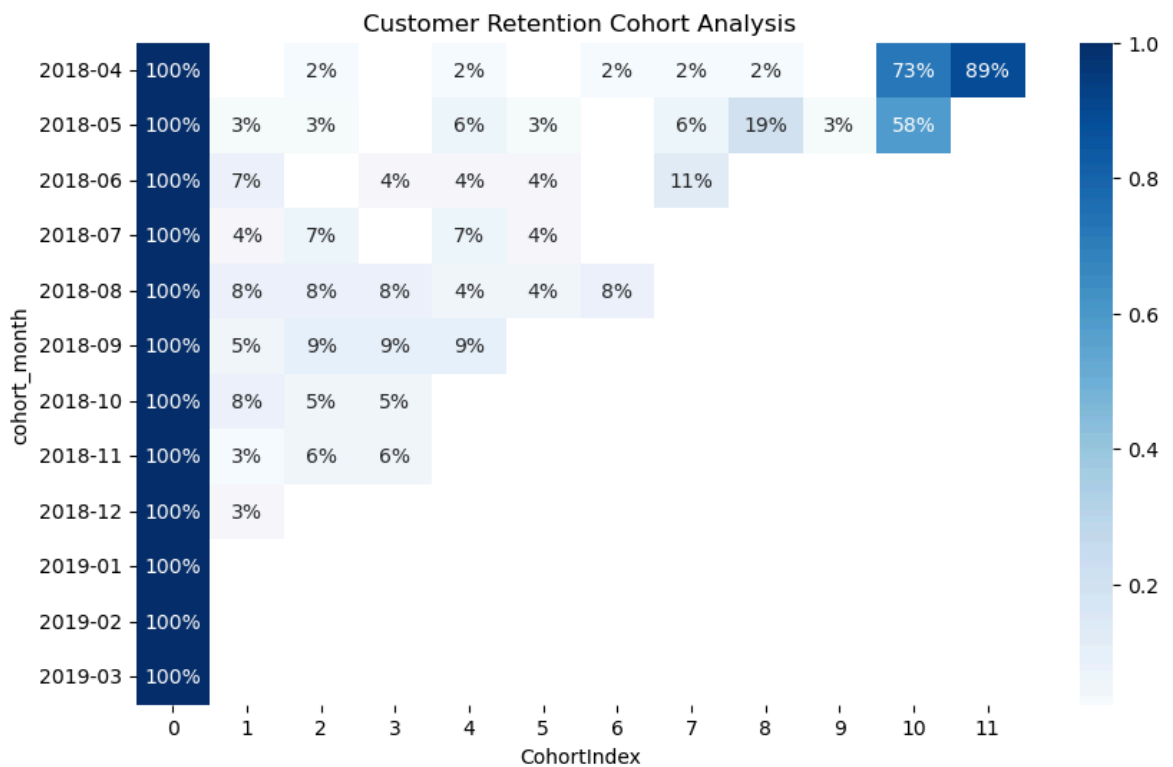
```
In [57]: cohort_data = df.groupby(['cohort_month', 'CohortIndex'])['Customer_id'].nunique
```

```
In [59]: cohort_pivot = cohort_data.pivot(index='cohort_month', columns='CohortIndex', va
```

```
In [61]: cohort_size = cohort_pivot.iloc[:, 0]

retention_matrix = cohort_pivot.divide(cohort_size, axis=0)
```

```
In [63]: plt.figure(figsize=(10,6))
sns.heatmap(retention_matrix, annot=True, fmt=".0%", cmap="Blues")
plt.title("Customer Retention Cohort Analysis")
plt.show()
```



```
In [70]: df.to_csv("CleanedData.csv", index=False)
retention_matrix.to_csv("CohortRetention.csv")
```

```
In [ ]:
```